

# Churn Analysis and Customer Intelligence

Sohail Khan | Data analysis project on an OTT subscription platform

---

## Objective

Find out which subscribers leave, what they have in common, and when the losses happened, so the retention team knows where to act first.

## Data and method

I pulled three tables (customers, subscriptions, support) from a SQLite database into Python with sqlite3 and pandas. The base has 21 customers, 6 of whom cancelled.

- **Cleaning:** renamed columns, converted dates, unified gender labels (Men/Women to Male/Female), dropped empty columns (interests, pincode, comment), and filled 3 missing countries by matching each customer's state to a known country.
- **Support data:** counted complaints per customer, then kept each customer's latest complaint so the join stayed one row per customer.
- **Features:** a churn flag (a cancellation date exists), tenure in days, and a risk tier from the churn score: low below 50, medium 50 to 69, high 70 and above.
- **Analysis and charts:** group-bys, a pivot table, and a correlation heatmap in pandas, matplotlib and seaborn.

## Headline numbers

**28.6%**

Churn rate (6 of 21)

**71.4%**

Retention rate

**18.85**

ARPU (monthly charges per customer)

**1,541 days**

Average tenure

**73.94**

Monthly revenue lost to churn

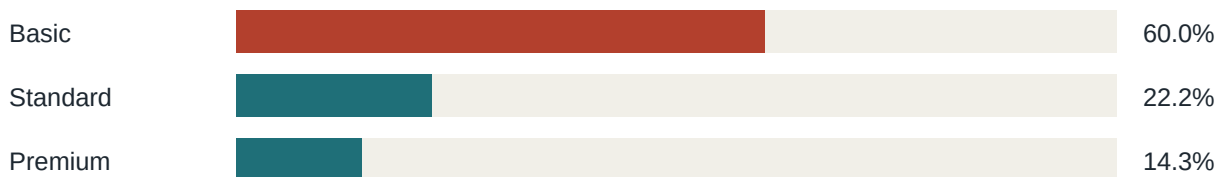
**18.7%**

Share of monthly revenue lost

Total monthly revenue is 395.79 across all 21 customers. Tenure counts to the cancellation date for churned customers and to the run date for active ones.

## Churn by plan

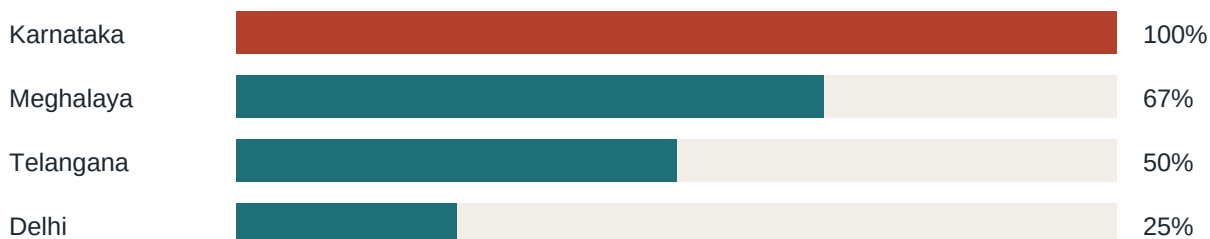
Basic customers churn at more than four times the Premium rate. Basic is also the smallest plan by revenue, so the losses there are painful in customer numbers but light in money.



| Plan     | Customers | Monthly revenue | Churn rate |
|----------|-----------|-----------------|------------|
| Basic    | 5         | 52.95           | 60.0%      |
| Standard | 9         | 123.91          | 22.2%      |
| Premium  | 7         | 218.93          | 14.3%      |

## Churn by state

Karnataka lost every customer in the data. Meghalaya, Telangana and Delhi also lost some, while five other locations lost none. Each state holds only a few customers, so these rates move a lot with a single cancellation.



## When customers left

Cancellations were spread over five months of 2024. September was the peak with 2 customers, and every other month had 1.

| Feb | May | Sep | Oct | Nov |
|-----|-----|-----|-----|-----|
| 1   | 1   | 2   | 1   | 1   |

## Support and escalations

Only 7 of the 21 customers have a support record, with 9 complaints in total (0.43 per customer). Four of those 7 customers had their latest complaint escalated, which is 19.1% of all customers. Among customers with a support record, escalation and churn were strongly correlated (0.77).

Treat that correlation as a lead, not a result. It rests on 7 customers, and a single different case would change it a lot.

## Recommendations

- **Contact high-risk customers first.** List everyone with a churn score of 70 or more, rank them by lifetime value, and reach out about their open issues.
- **Look into Karnataka.** Check for a price change, service problems or complaints there that could explain losing every customer.
- **Review the Basic plan and September 2024.** Check whether pricing or product changes came before the Basic losses and the September peak.
- **Act on escalated complaints quickly.** Escalated customers looked far more likely to leave, so closing those cases fast is a cheap retention step.
- **Read the cancellation reasons.** Reasons such as "Too expensive" and "Switched to competitor" point to pricing and competitor checks.

## Limitations

- The dataset is small (21 customers), so percentages by state, plan and month are indicative, not statistically firm.
- ARPU is calculated across all customers, including those who cancelled.
- Only each customer's latest complaint is kept in the joined table, so earlier escalations are not reflected in the correlation.